

Splitting one model
across devices

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graph TD; A[Splitting one model across devices] --> B[Horizontal partitioning]; A --> C[Vertical partitioning]; B --> D[Communication overhead: layer 1's output must be copied before layer 2 runs. GPUs sit idle.]; C --> E[matmul and convolution are highly parallelizable, so this is easy to implement and almost always the best fit.];
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Horizontal partitioning

each device handles
different layers

Communication overhead:
layer 1's output must be
copied before layer 2 runs.

GPUs sit idle.

Vertical partitioning

each layer split across
all devices

matmul and convolution are
highly parallelizable, so this
is
easy to implement and
almost always the best fit.